AWS Cloud Security Basics

Identity and Access Management

AWS IAM controls access to AWS resources. Policies define permissions, and roles can be assumed by users, services, or workloads. Production environments should follow the principle of least privilege.

Encryption

AWS services such as KMS provide managed cryptographic keys. Encryption can protect data at rest and can be combined with TLS to protect data in transit.

Secrets

Sensitive values such as passwords, API keys, and tokens should not be stored directly in source code. AWS Secrets Manager and Systems Manager Parameter Store can be used to manage configuration and secrets.

Network Security

VPC security groups act as stateful virtual firewalls for AWS resources. Network ACLs provide subnet-level stateless filtering. Private subnets can reduce direct exposure of internal workloads.

Monitoring

Amazon CloudWatch can collect metrics, logs, and alarms. Security monitoring should combine application logs, infrastructure telemetry, and AWS security services to detect abnormal activity.